Statistical Mechanics: Pressure, Free Energy, and Energy Fluctuations

Leonard Susskind

Lecture notes organized by LazyingArt LLC with Video2Book

Original lecture by Leonard Susskind. Lecture notes organized by LazyingArt LLC with Video2Book.

Chapter 1

Pressure, Free Energy, and Energy Fluctuations

Overview. We derive pressure from the partition function without assuming a kinetic picture of molecules striking a wall. A geometric lemma converts an adiabatic derivative into fixed-temperature derivatives, the Helmholtz free energy supplies the bridge to $\log Z$, and the ideal-gas law follows as the first application. We then use the same partition function to derive the equilibrium energy fluctuation and its relation to heat capacity.

1.1 Let the Mathematics Test the Intuition

Thermodynamics has a peculiar reputation. Its calculus is elementary, yet the choice of what to differentiate and what to hold fixed can be remarkably subtle. Our problem is deliberately narrow: calculate the pressure of an ideal gas. The route to that answer, however, is intended to be general enough for systems in which the familiar molecular picture is no longer reliable.

The kinetic story is easy to tell. Molecules strike a wall and reverse the normal component of their momentum. Momentum delivered per unit time gives the force, and force per unit area gives the pressure. We already know that the mean kinetic energy per molecule in three dimensions is

$$\left\langle \frac{p^2}{2m} \right\rangle = \frac{3}{2}T \qquad (k_B = 1). \tag{1.1}$$

With an isotropic velocity distribution one can estimate the collision rate, the momentum transfer in each collision, and the resulting pressure.

But the wall itself seems to spoil isotropy. Why should molecules very near the boundary have the same distribution of speeds and directions as molecules deep in the interior? In equilibrium they do, to the accuracy relevant here, but that conclusion should follow from statistical mechanics rather than be smuggled into the calculation as intuition. Collisions among the molecules make the kinetic argument still less transparent. The noninteracting gas is easy precisely because it is an unusually simple system.

There are occasions in physics when the safest procedure is to suppress the picture temporarily and ride the mathematics. One identifies the quantity to be found, manipulates the general relations, and waits until a recognizable formula appears. Intuition returns at the end, but then the mathematics is testing the intuition rather than the other way around. The pleasure of thermodynamics lies in the surprising relations that emerge from this procedure.

The desired relation is an equation of state,

$$P = P(T, V, N), \tag{1.2}$$

where N is the amount of material, T the temperature, and V the volume. A satisfactory derivation should continue to make sense when the particles interact, or when the material is a liquid, plasma, or solid. Only the final evaluation of the partition function will specialize to an ideal gas.

1.2 Entropy and the Helmholtz Free Energy

Begin by retrieving the canonical formulas that will be needed later. For microstates labelled by i,

$$\begin{aligned} S &= -\sum_i p_i \log p_i, \\ p_i &= \frac{e^{-\beta E_i}}{Z}, \\ Z &= \sum_i e^{-\beta E_i}. \end{aligned} \tag{1.3}$$

The logarithm of the probability is

$$\log p_i = -\beta E_i - \log Z. \tag{1.4}$$

Substitution into the entropy gives

$$\begin{aligned} S &= -\sum_i p_i \log p_i \\ &= \beta \sum_i p_i E_i + \log Z \sum_i p_i \\ &= \beta E + \log Z. \end{aligned} \tag{1.5}$$

Here $E = \sum_i p_i E_i$ is the mean energy.

> **Question from the audience.** Why does the second term give only $\log Z$, rather than the number of states times $\log Z$?
>
> **Response.** The coefficient being summed is p_i, not 1. Normalization gives $\sum_i p_i = 1$. Equivalently, the numerator $\sum_i e^{-\beta E_i} = Z$ cancels the Z in the denominator. The number of states never appears in this step. [1]

In the energy units used through most of the lecture, $\beta = 1/T$. Multiplying (1.5) by T and rearranging gives

$$E - TS = -T \log Z. \tag{1.6}$$

The combination on the left occurs so often that it deserves a name:

$$A \equiv E - TS, \qquad A = -T \log Z. \tag{1.7}$$

It is the Helmholtz free energy. The name is less important at this moment than the identity itself; we shall need the identity when pressure is converted into a derivative of the partition function.

[1] Lecture timestamp: 00:14:25–00:14:55.

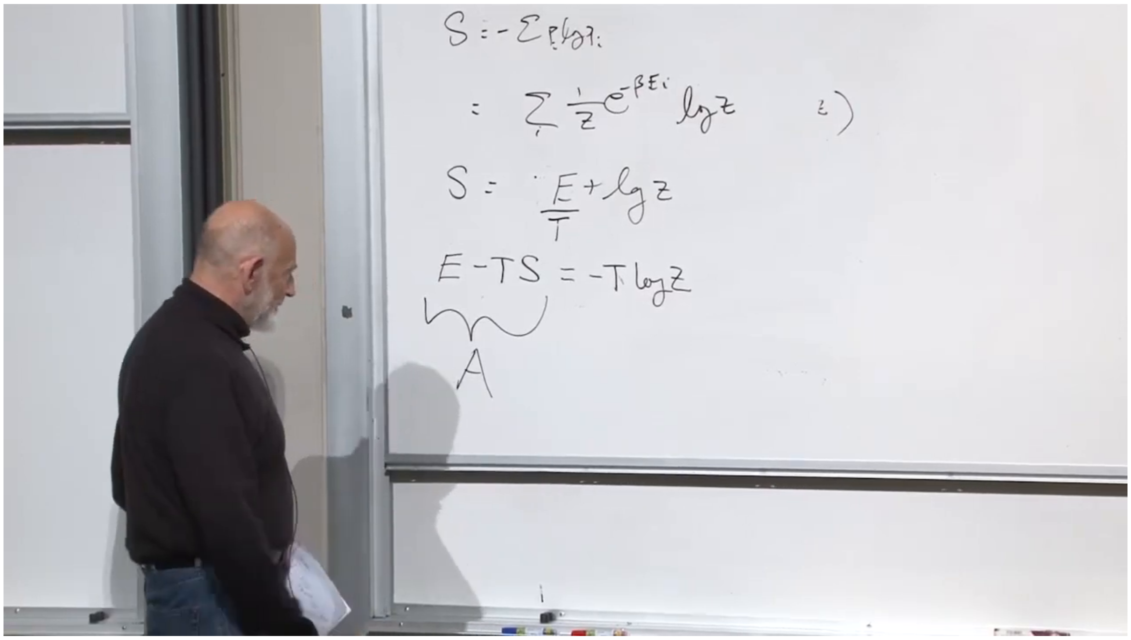

Figure 1.1: The canonical entropy relation reorganized into $A = E - TS = -T \log Z$.
Lecture frame: 00:16:50.

1.3 Control Parameters and Their Responses

A control parameter is a macroscopic quantity that an experimenter can change from outside. Moving a piston changes the volume without manipulating individual molecules. Applied electric and magnetic fields are other examples. In a more general ensemble, the number of particles can also be allowed to change.

Thermodynamic variables naturally come in conjugate pairs. Volume is the control parameter in the present problem, and pressure is its response. We change V; the pressure tells us how the system energetically resists that change. Later the derivative of the energy will make this pairing precise.

Question from the audience. Can pressure itself be regarded as the control parameter?

Response. One may design an experiment that fixes the external pressure, but in the construction used here the piston position fixes the volume and the internal pressure is the response. Pressure and volume are conjugate variables; which member is controlled depends on the ensemble and apparatus. [2]

Question from the audience. Are all the molecules assumed to be identical, or is the argument restricted to a gas?

Response. No such restriction is needed for the thermodynamic argument. The cylinder may contain several species, a liquid, or even a solid being compressed. The ideal-gas partition function will be a special application only after the general pressure formula has been established. [3]

[2] Lecture timestamp: 00:19:22–00:20:01.
[3] Lecture timestamp: 00:20:02–00:20:39.

Question from the audience. Could the amount of material be changed as another external operation?

Response. Yes. Particle number is another possible control parameter. Treating it dynamically leads to its own conjugate variable, the chemical potential. For the calculation tonight, N remains fixed. [4]

Question from the audience. Can liquid water and steam be present simultaneously in equilibrium?

Response. At the coexistence temperature and pressure they can. Gravity may place the liquid below the vapor, yet the two phases can still share one temperature and remain in thermal equilibrium. The same general statistical-mechanical framework applies. [5]

For the next argument, regard energy and entropy as dependent functions of two independently adjustable variables:

$$E = E(T, V), \qquad S = S(T, V). \tag{1.8}$$

Temperature and volume provide coordinates on a thermodynamic state space. Energy and entropy assign numbers to every equilibrium point in that space.

1.4 The Constant-Entropy Derivative

The derivative we ultimately need is unusual:

$$\left(\frac{\partial E}{\partial V}\right)_S .$$

Although T and V are the independent coordinates, this derivative asks us to change V while constraining the dependent function $S(T, V)$ to remain constant. It is a derivative along a curve in the (T, V)-plane, not along a coordinate axis.

Question from the audience. How can T and V be independent if a derivative dT/dV is later used?

Response. They are independent on the full state space. Once we impose the constraint $S(T, V) =$ constant, the allowed states lie on a one-dimensional contour. Along that contour, T becomes a function of V, and its slope dT/dV is well defined. [6]

Lemma 1.1. *If $E(T, V)$ and $S(T, V)$ are differentiable, then wherever $(\partial S/\partial T)_V \neq 0$,*

$$\left(\frac{\partial E}{\partial V}\right)_S = \left(\frac{\partial E}{\partial V}\right)_T - \left(\frac{\partial E}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T . \tag{1.9}$$

[4] Lecture timestamp: 00:20:39–00:20:57.
[5] Lecture timestamp: 00:20:59–00:21:36.
[6] Lecture timestamp: 01:01:18–01:02:09.

Proof. For an arbitrary infinitesimal displacement,

$$dE = \left(\frac{\partial E}{\partial V}\right)_T dV + \left(\frac{\partial E}{\partial T}\right)_V dT. \tag{1.10}$$

At fixed V, the chain rule permits S to be used as an intermediate variable:

$$\left(\frac{\partial E}{\partial T}\right)_V = \left(\frac{\partial E}{\partial S}\right)_V \left(\frac{\partial S}{\partial T}\right)_V . \tag{1.11}$$

Now restrict the displacement to a contour on which S is constant. Along that contour,

$$0 = dS = \left(\frac{\partial S}{\partial V}\right)_T dV + \left(\frac{\partial S}{\partial T}\right)_V dT. \tag{1.12}$$

Solving for the contour slope gives

$$\left(\frac{dT}{dV}\right)_S = -\frac{\left(\frac{\partial S}{\partial V}\right)_T}{\left(\frac{\partial S}{\partial T}\right)_V}. \tag{1.13}$$

Divide (1.10) by dV, use (1.11), and insert (1.13):

$$\begin{aligned}\left(\frac{\partial E}{\partial V}\right)_S &= \left(\frac{\partial E}{\partial V}\right)_T + \left(\frac{\partial E}{\partial S}\right)_V \left(\frac{\partial S}{\partial T}\right)_V \left(\frac{dT}{dV}\right)_S \\ &= \left(\frac{\partial E}{\partial V}\right)_T - \left(\frac{\partial E}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T . \end{aligned} \tag{1.14}$$

The factor $(\partial S/\partial T)_V$ has cancelled, proving the result. □

The geometric picture is not ornamental. It explains why an apparently strange fixed-S derivative can be evaluated with ordinary partial derivatives in the independent coordinates T and V. Writing $\delta E/\delta V$ during the construction is a useful reminder that the symbols represent ratios of small changes along a specified path.

Question from the audience. Is the expansion of dE just the ordinary definition of partial derivatives?

Response. Yes. Nothing special occurs in the total differential. The only extra ingredient is the later restriction to the contour $dS = 0$, which determines the relation between dT and dV. [7]

Question from the audience. Does the cancellation require $(\partial S/\partial T)_V$ to be nonzero?

Response. Yes. For an ordinary stable system,

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{C_V}{T} > 0,$$

so the division is legitimate. Exceptional points require separate care. At fixed Hamiltonian, raising the temperature broadens the canonical energy distribution and normally increases the entropy; lowering the temperature narrows it. This is the physically correct orientation of the monotonicity argument. [8]

[7] Lecture timestamp: 00:29:26–00:29:32.
[8] Lecture timestamp: 00:36:50–00:37:47.

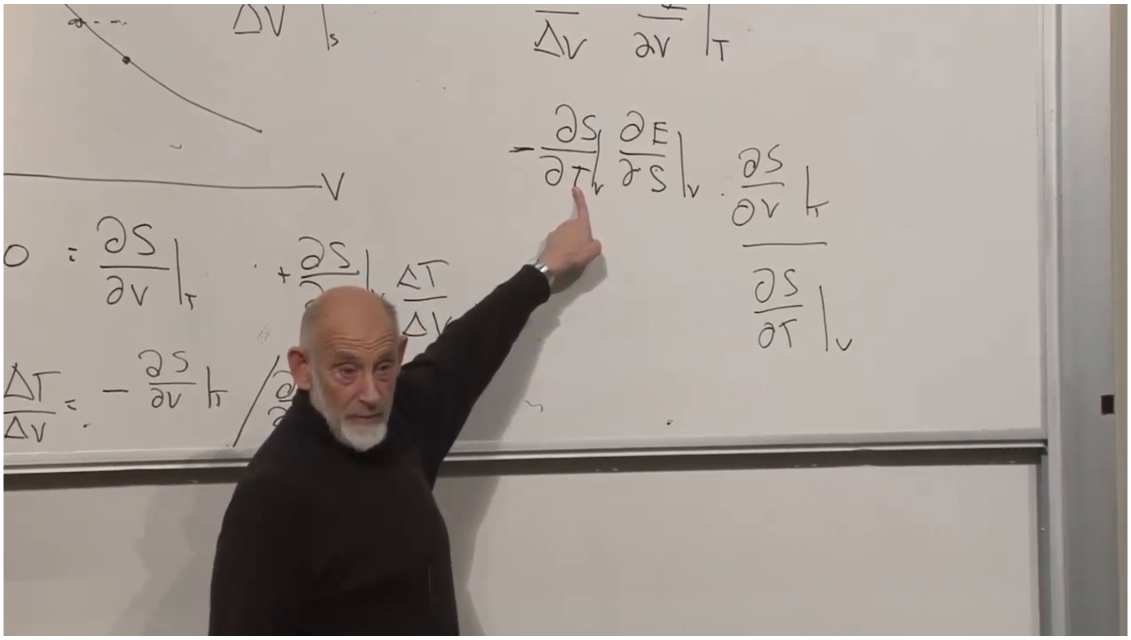

Figure 1.2: The constant-entropy contour calculation at the cancellation that completes the derivative lemma.

Lecture frame: 00:36:10.

1.5 Piston Work and the Adiabatic Condition

We now give the derivative a physical meaning. Consider a piston with cross-sectional area $\mathcal{A}$. If it moves outward by dx, then

$$dV = \mathcal{A}\,dx. \tag{1.15}$$

The gas exerts force $P\mathcal{A}$ on the piston, so the work done by the gas is

$$\delta W_{\text{by}} = P\mathcal{A}\,dx = P\,dV. \tag{1.16}$$

If no heat enters the gas, energy conservation gives

$$dE = -P\,dV. \tag{1.17}$$

The sign is physical: during expansion the gas raises or accelerates the piston, and its own energy decreases.

Two qualifications are essential. The piston must move slowly enough for the system to remain arbitrarily close to equilibrium, and the walls must be insulated so that energy is not supplied as heat. A rapid pull can be so fast that no molecule reaches the receding piston during the motion. A rapid compression launches shock waves. Neither process is represented by a sequence of equilibrium states.

The word *adiabatic* is sometimes used for no heat transfer and sometimes for slow change. In the reversible process required here, both ideas enter: the change is quasistatic and thermally insulated. Under those conditions it is also isentropic.

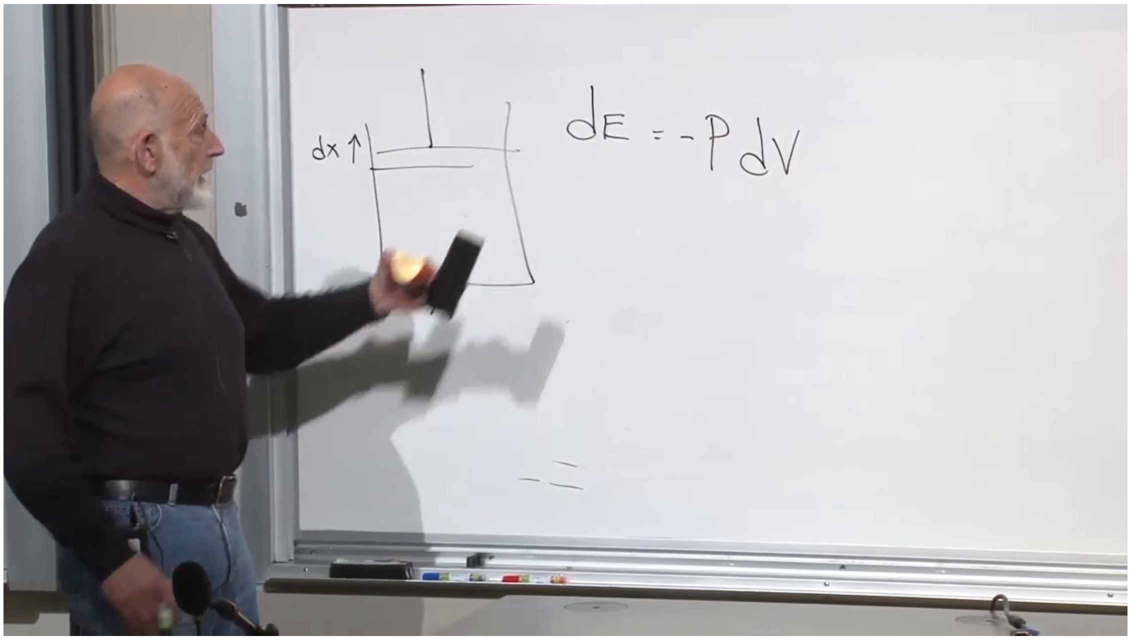

Figure 1.3: The piston geometry and the completed work relation $dE = -P\,dV$.
Lecture frame: 00:43:10.

Question from the audience. Why does an adiabatic change keep the entropy fixed?

Response. View the system quantum mechanically. Its many-body energy levels depend on the volume. When V changes slowly, the adiabatic theorem carries each occupied level continuously into its successor without transitions between level labels. The energies $E_i(V)$ move, but the occupation probabilities p_i remain fixed. Since $S = -\sum_i p_i \log p_i$ depends on those probabilities, the entropy remains constant. Heat exchange or irreversible motion would invalidate this conclusion. [9]

Equation (1.17), evaluated along this constant-entropy path, identifies pressure as

$$P = -\left(\frac{\partial E}{\partial V}\right)_S . \tag{1.18}$$

This is the conjugate relation between pressure and volume.

Question from the audience. What happens if the piston is moved rapidly instead?

Response. The system leaves equilibrium and its detailed dynamics matter. Sudden compression generates a shock. In an ideal free expansion, the piston can recede before any molecule strikes it, so no work is done and the energy remains fixed even though the volume changes. The gas later relaxes to equilibrium with increased entropy. Such processes cannot be described by the single equilibrium derivative in (1.18). [10]

[9] Lecture timestamp: 00:43:37–00:49:15.
[10] Lecture timestamp: 00:58:30–01:01:15.

Figure 1.4: Many-body energy levels moving with the volume while their occupation probabilities retain their level labels.

Lecture frame: 00:48:50.

Question from the audience. What quantitative condition decides whether a change is slow?

Response. The external variation time must be long compared with the relevant internal relaxation times. For a gas, many molecular collisions and many wall impacts should occur during the macroscopic change. The precise time scale depends on the material and geometry, while the quasistatic limit itself is general. [11]

1.6 From Adiabatic Energy to the Partition Function

The physical formula (1.18) is not yet the easy one to calculate. Canonical statistical mechanics naturally supplies functions of temperature, because T appears directly in the Boltzmann weight. The calculus lemma converts the awkward fixed-entropy derivative into that calculational language:

$$P = -\left(\frac{\partial E}{\partial V}\right)_T + \left(\frac{\partial E}{\partial S}\right)_V \left(\frac{\partial S}{\partial V}\right)_T . \tag{1.19}$$

At fixed volume the first law gives $dE = T\,dS$, and therefore

$$\left(\frac{\partial E}{\partial S}\right)_V = T. \tag{1.20}$$

Thus

$$P = -\left(\frac{\partial E}{\partial V}\right)_T + T\left(\frac{\partial S}{\partial V}\right)_T . \tag{1.21}$$

[11] Lecture timestamp: 01:02:09–01:03:20.

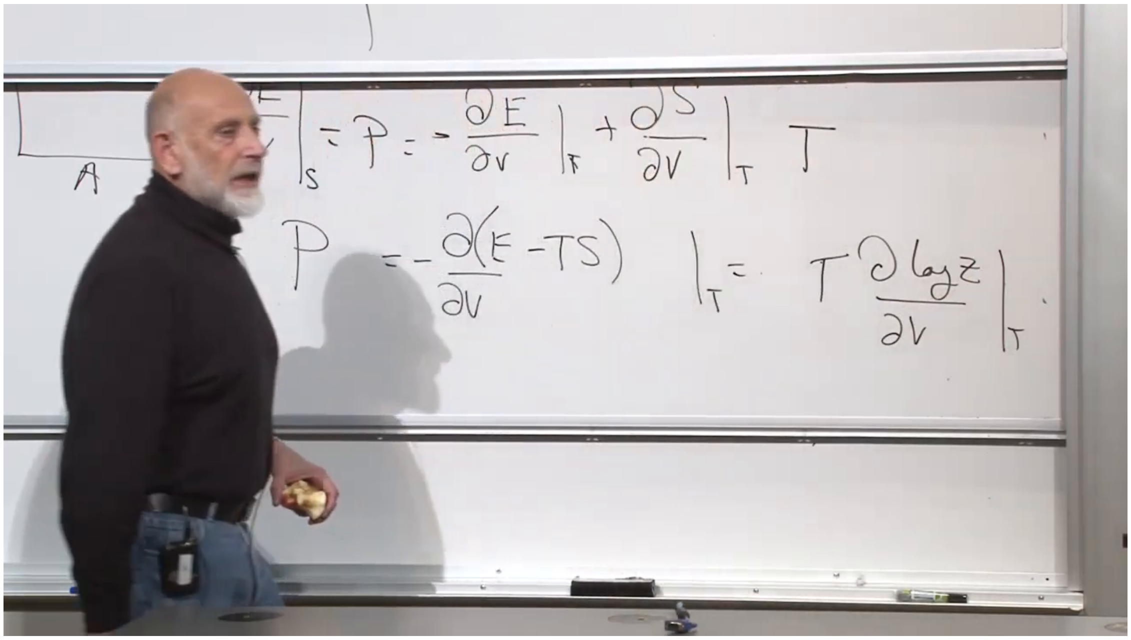

Figure 1.5: The fixed-temperature compression of the argument: $P = -(\partial A/\partial V)_T = T(\partial \log Z/\partial V)_T$.

Lecture frame: 00:55:55.

Question from the audience. Where does the minus sign in the pressure formula come from?

Response. It originates in work done by the gas: $dE = -P\,dV$ during adiabatic expansion. The calculus lemma itself has a minus sign between its two terms; the overall minus in $P = -(\partial E/\partial V)_S$ reverses both of them, producing (1.21). [12]

Because T is held fixed in both derivatives,

$$
\begin{aligned}
P &= -\left(\frac{\partial}{\partial V}(E - TS)\right)_T \\
&= -\left(\frac{\partial A}{\partial V}\right)_T .
\end{aligned}
\tag{1.22}
$$

Finally insert $A = -T\log Z$:

$$
\boxed{P = T\left(\frac{\partial \log Z}{\partial V}\right)_T}. \tag{1.23}
$$

Nothing in (1.23) assumes an ideal gas. If the partition function can be calculated for an interacting fluid, plasma, or solid, the same derivative gives its pressure. More generally, every external control parameter has a conjugate response obtained by differentiating an appropriate thermodynamic potential. The partition function packages many such responses: differentiating $\log Z$ with respect to β gives energy, while differentiating it with respect to volume gives pressure.

[12] Lecture timestamp: 00:52:40–00:53:14.

1.7 The Ideal-Gas Equation of State

Only now specialize to N noninteracting particles. With the classical phase-space normalization retained,

$$Z_N = \frac{1}{N!h^{3N}} \int d^{3N}q\, d^{3N}p\, \exp\left[-\beta \sum_{a=1}^{N} \frac{\mathbf{p}_a^2}{2m}\right]. \tag{1.24}$$

There are $3N$ position coordinates and $3N$ momentum coordinates. The Hamiltonian is independent of position, so every particle contributes one factor of the container volume:

$$Z_N = \frac{V^N}{N!h^{3N}} I_p(\beta), \tag{1.25}$$

where the entire momentum integral $I_p(\beta)$ is independent of V.

> **Question from the audience.** Why are there $3N$ position integrals and $3N$ momentum integrals?
>
> **Response.** Each of the N particles has three spatial coordinates and three conjugate momentum components. The count is $3N$ in each half of phase space, not N particles plus another $3N$ particles. [13]

Taking a logarithm separates the volume dependence:

$$\log Z_N = N \log V - \log N! - 3N \log h + \log I_p(\beta). \tag{1.26}$$

The factors $1/N!$, h^{-3N}, and $I_p(\beta)$ are all independent of volume. They matter for absolute entropy and other questions, but not for this particular derivative. Hence

$$\left(\frac{\partial \log Z_N}{\partial V}\right)_T = \frac{N}{V}. \tag{1.27}$$

Equation (1.23) immediately gives

$$P = \frac{NT}{V}, \qquad \boxed{PV = NT}. \tag{1.28}$$

> **Question from the audience.** Does N still count the replicas used to derive the Boltzmann distribution?
>
> **Response.** No. The replica construction was a counting device in the earlier derivation and is no longer present. Here N is the number of physical particles in this one gas sample. [14]

Defining the number density

$$\rho = \frac{N}{V} \tag{1.29}$$

gives the equivalent form

$$P = \rho T. \tag{1.30}$$

[13] Lecture timestamp: 01:04:19–01:04:28.
[14] Lecture timestamp: 01:07:38–01:08:05.

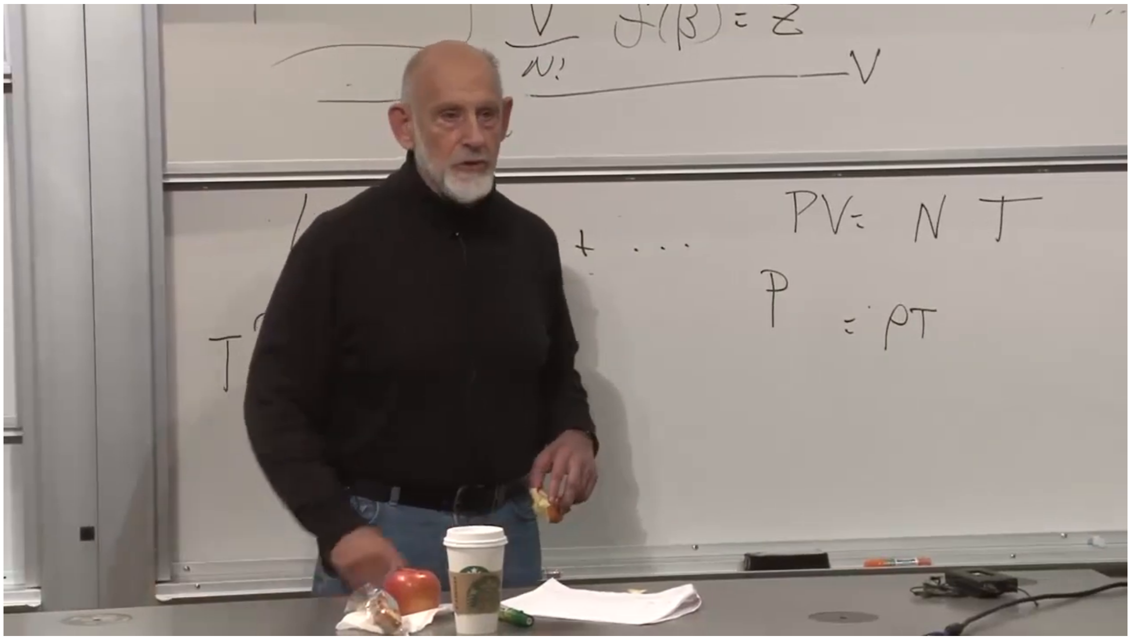

Figure 1.6: The volume derivative of $\log Z$ and the resulting ideal-gas equation of state $PV = NT$, or $P = \rho T$.

Lecture frame: 01:09:25.

Question from the audience. Should the density form be obtained by dividing by N or by V?

Response. Divide $PV = NT$ by V: $P = (N/V)T = \rho T$. Here ρ is number density, not mass density. [15]

Restoring laboratory temperature T_K,

$$PV = Nk_BT_K. \tag{1.31}$$

The familiar result agrees with kinetic intuition, but its derivation now rests on a framework that survives far beyond the free gas.

1.8 A Second Dividend: Equilibrium Fluctuations

The pressure calculation is complete, but one promised question remains: what else can be extracted from the partition function? Consider a system in thermal contact with a heat bath. Its energy is not fixed. Energy enters and leaves, so repeated measurements fluctuate around the canonical mean.

Question from the audience. Can we now calculate the fluctuations that were promised?

Response. Yes. The extra time allows a second demonstration of the same principle: derivatives of Z encode measurable thermodynamic information. [16]

[15] Lecture timestamp: 01:08:45–01:09:16.
[16] Lecture timestamp: 01:10:37–01:10:50.

1.8.1 Variance and Standard Deviation

For a random variable x with $\langle x\rangle = 0$, a measure of the distribution's width is $\langle x^2\rangle$. For a general distribution, shift the variable by its mean:

$$\delta x = x - \langle x\rangle. \tag{1.32}$$

Its mean vanishes,

$$\langle \delta x\rangle = \langle x\rangle - \langle x\rangle = 0, \tag{1.33}$$

and its mean square is

$$\begin{aligned}(\Delta x)^2 &\equiv \left\langle (x - \langle x\rangle)^2 \right\rangle \\ &= \langle x^2\rangle - 2\langle x\rangle^2 + \langle x\rangle^2 \\ &= \langle x^2\rangle - \langle x\rangle^2. \end{aligned} \tag{1.34}$$

This is the variance. Its positive square root Δx is the standard deviation.

Question from the audience. Why is the average of $x - \langle x\rangle$ exactly zero?

Response. Averages are linear, and $\langle x\rangle$ is already a number. Therefore $\langle x - \langle x\rangle\rangle = \langle x\rangle - \langle x\rangle = 0$. [17]

Question from the audience. What is the distinction among squared fluctuation, variance, root-mean-square width, and standard deviation?

Response. The variance is $(\Delta x)^2 = \langle x^2\rangle - \langle x\rangle^2$. The standard deviation is its square root, Δx. For a zero-mean variable the standard deviation is also its root-mean-square value. [18]

1.8.2 Energy Moments from the Partition Function

For the canonical ensemble,

$$\begin{aligned}\langle E\rangle &= \frac{1}{Z}\sum_i E_i e^{-\beta E_i} \\ &= -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = -\frac{\partial \log Z}{\partial \beta}. \end{aligned} \tag{1.35}$$

Multiplication by one factor of E_i is produced by one negative β-derivative. A second derivative generates E_i^2:

$$\langle E^2\rangle = \frac{1}{Z}\sum_i E_i^2 e^{-\beta E_i} = \frac{1}{Z}\frac{\partial^2 Z}{\partial \beta^2}. \tag{1.36}$$

[17] Lecture timestamp: 01:14:02–01:14:22.
[18] Lecture timestamp: 01:16:39–01:18:38.

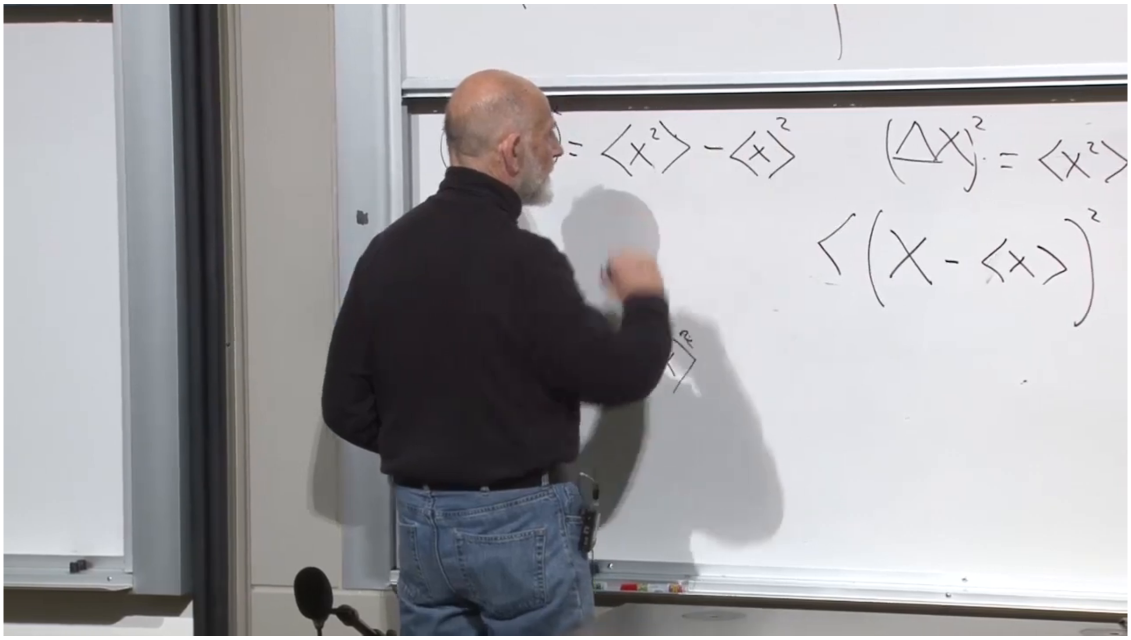

Figure 1.7: The shifted variable and the expansion that produces $(\Delta x)^2 = \langle x^2\rangle - \langle x\rangle^2$.
Lecture frame: 01:17:20.

The energy variance is consequently

$$\begin{aligned}(\Delta E)^2 &= \langle E^2\rangle - \langle E\rangle^2\\ &= \frac{1}{Z}\frac{\partial^2 Z}{\partial\beta^2} - \frac{1}{Z^2}\left(\frac{\partial Z}{\partial\beta}\right)^2\\ &= \frac{\partial^2 \log Z}{\partial\beta^2} = -\frac{\partial\langle E\rangle}{\partial\beta}.\end{aligned} \tag{1.37}$$

Once again, a complicated-looking observable becomes a derivative of $\log Z$.

1.8.3 Heat Capacity and the Size of Fluctuations

In lecture units $T = 1/\beta$, so

$$-\frac{\partial}{\partial\beta} = T^2\frac{\partial}{\partial T}. \tag{1.38}$$

At fixed volume define the heat capacity of the whole sample by

$$C_V \equiv \left(\frac{\partial\langle E\rangle}{\partial T}\right)_V. \tag{1.39}$$

Equation (1.37) becomes

$$\boxed{(\Delta E)^2 = T^2 C_V} \qquad (k_B = 1). \tag{1.40}$$

For laboratory temperature T_K, where $\beta = (k_B T_K)^{-1}$,

$$\boxed{(\Delta E)^2 = k_B T_K^2 C_V}. \tag{1.41}$$

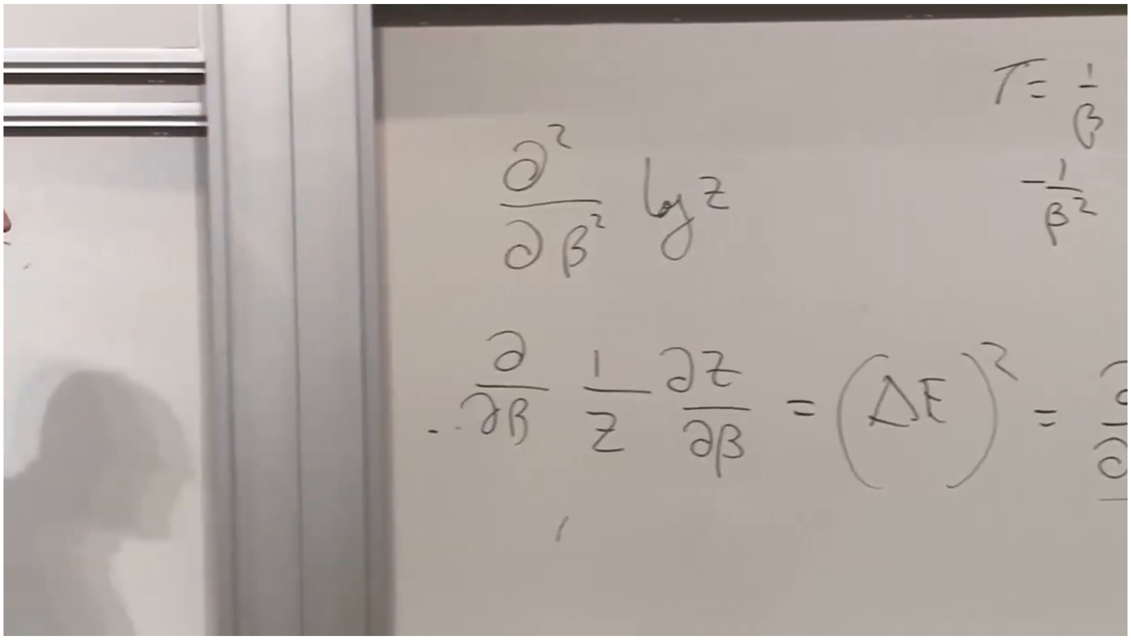

Figure 1.8: The second β-derivative of $\log Z$ identified with the energy variance.
Lecture frame: 01:29:00.

Question from the audience. Must the derivative specify fixed volume rather than fixed pressure?

Response. Yes. This canonical derivation keeps the Hamiltonian's external control parameters fixed, including V. The resulting quantity is C_V. A constant-pressure ensemble allows volume and work to fluctuate and requires the corresponding thermodynamic potential; one cannot simply replace C_V by C_P in (1.41). [19]

Question from the audience. Is C_V a material constant?

Response. Not in general. It may depend on temperature, volume, phase, and composition. Moreover, C_V here is the heat capacity of the entire sample and is extensive. A specific heat is obtained only after dividing by mass, particle number, or amount of substance. [20]

For ordinary extensive matter, C_V is proportional to N, so $(\Delta E)^2$ is proportional to N and ΔE to $\sqrt{N}$. The mean energy itself is proportional to N. Therefore

$$\frac{\Delta E}{\langle E\rangle} \sim \frac{1}{\sqrt{N}}. \tag{1.42}$$

Macroscopic relative fluctuations are tiny, while microscopic samples display them directly. The appearance of k_B in (1.41) marks this connection between thermodynamic observables and molecular discreteness; equivalently, it connects macroscopic units to Avogadro-scale counting.

[19] Lecture timestamp: 01:29:01–01:29:28; 01:32:50–01:33:21.
[20] Lecture timestamp: 01:31:47–01:32:21; 01:33:23–01:34:55.

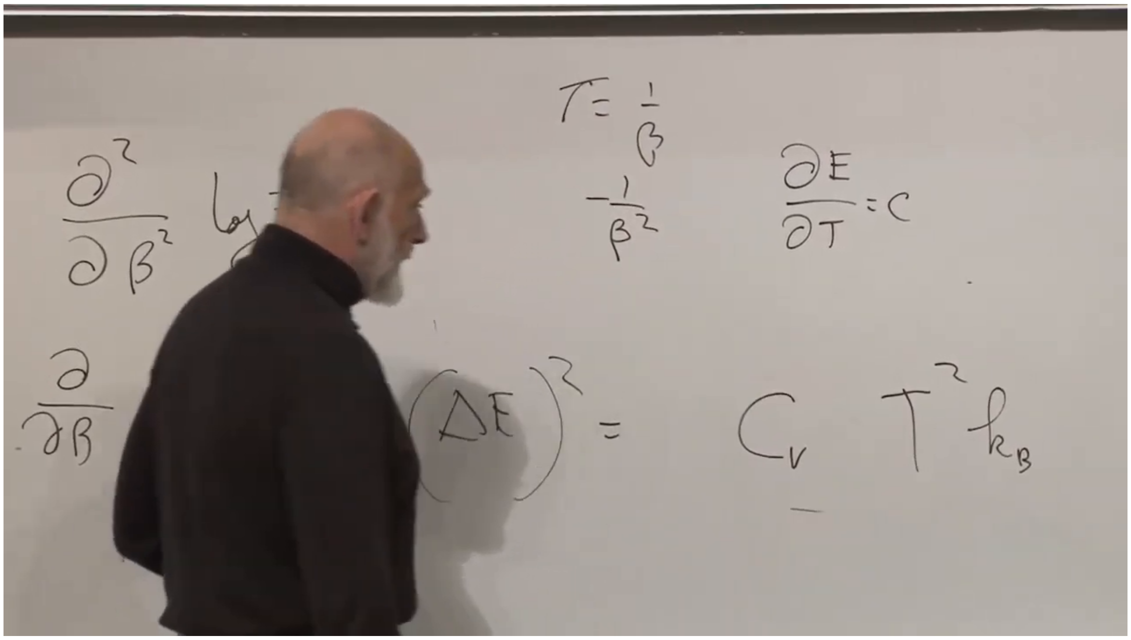

Figure 1.9: The conversion from the β-derivative to the fixed-volume heat capacity, including the restored factor of k_B.

Lecture frame: 01:31:10.

Pressure and fluctuations have now emerged from the same object:

$$P = T\left(\frac{\partial \log Z}{\partial V}\right)_T, \qquad (\Delta E)^2 = \frac{\partial^2 \log Z}{\partial \beta^2}.$$

This is the central lesson. Once the correct constraints are identified, the partition function does not merely normalize probabilities. Its derivatives organize the measurable thermodynamics of the system.